Enguerrand LE NILIOT

Seoul, South Korea (Available for a 6-month Engineering Internship - Starting Jan 2027)
enguerrand.leniliot@gmail.com — +82 010-9572-3715
linkedin.com/in/enguerrand-le-niliot — github.com/EnguerrandLN — enguerrand-leniliot.com

Education

Seoul National University (SNU) Seoul, South Korea
Academic Exchange Program Sep 2026 – Dec 2026
- **Focus:** Graduate coursework in Artificial Intelligence and Applied Mathematics.

Télécom Paris (Institut Polytechnique de Paris) Paris, France
MSc in Engineering (Top-tier French Engineering Institution) Sep 2024 – Expected Jun 2027
- **Double Major:** MACS (Random Modeling & Scientific Computing) & IMA (Computer Vision & 3D).
- **Specialization Track:** Data Science.

Lycée Thiers (CPGE MPSI / MP*) Marseille, France
Intensive Preparation for French Grandes Écoles Entrance Exams Sep 2021 – Jun 2024
- Selective 3-year track with intensive curriculum in Mathematics, Physics, and Computer Science.

Technical Skills

Languages & Frameworks: Python, PyTorch, NumPy, SQL, GDScript.

Computer Vision & Deep Learning: 3D Vision (Point Clouds, Stereo), Tracking (Kalman, Optical Flow), Generative Models (GANs, VAEs), Medical Imaging, CNNs.

Applied Math & Stochastic Modeling: Stochastic Calculus (Brownian Motion, Martingales), Markov Chains, High-Dim Probability, Monte Carlo, Poisson Processes.

Data Science & Optimization: Statistical Learning (SVM, Random Forests, PCA), Algorithmic Optimization (Gradient Descent, Simplex), Estimation & Hypothesis Testing.

Featured Academic Projects

Computer Vision Engineer — MedTech Image Classification *Télécom Paris, 2025*
Deep Learning pipeline for 13-class leukocyte classification · [GitHub Repo]
- **End-to-End Deep Learning:** Engineered a classification pipeline for noisy microscopic images with severe class imbalance, achieving 0.7839 F1-Score Macro (**Top 5/77 ranking**).
- **Architecture & Optimization:** Implemented ConvNeXt utilizing Transfer Learning, AMP, and CLAHE preprocessing for clinical color standardization.
- **Advanced Regularization:** Mitigated data asymmetry and Domain Shift using WeightedRandomSampler, Focal Loss, and 4-Fold Stratified Ensembling (Soft Voting).

Machine Learning Engineer — Unsupervised Abstract Art Clustering *Télécom Paris, 2025*
Multi-modal feature extraction and clustering pipeline (Group Project) · [GitHub Repo]
- **CNN Feature Extraction:** Built a texture extraction engine using customized VGG16, computing Gram matrices across 4 layers with patch sampling and PCA.
- **Algorithmic Fusion & Clustering:** Fused CNN, VAE, and Geometric spaces via alpha-blending. Developed a "Core Silhouette" algorithm to prune ambiguous K-Means clusters.
- **Data Pipeline & Analysis:** Built an end-to-end evaluation pipeline featuring t-SNE reduction, cosine distance matrices, and automated DB scripts for a web interface.

Leadership & Extracurricular Activities

President of the Gaming Club — *Télécom Paris* Sep 2025 – Jun 2026
- Organized an inter-school LAN tournament with a **€3,000 cash prize**; managed budget, sponsorships, and logistics.
- Directed club administration and organized livestreamed (OBS) esports events.

Treasurer, Salon du Terroir — *Télécom Paris* Sep 2025 – Jun 2026
- Oversaw finances and logistics for a student exhibition drawing **over 1,000 attendees**.
- Built balance sheets tracking receipts, expenses, and treasury status for the year.

Communications Manager of the Sports Bureau (BDS) — *Télécom Paris* Sep 2025 – Jun 2026
- Led digital communication strategy and multimedia content for school-wide athletic events.

Languages & Interests

Languages: French (Native), English (Fluent, C1 - Linguaskill), Japanese (Intermediate, A2).

Interests: eSports, Video Game Development (Gameplay programming on Godot Engine), Scuba Diving, Traveling.